A2-LEVEL PHYSICS Sir AAMIR YASIN 03355000077

6-8-22 Sunday

Invasive Method

A process in which we study the structure, function or abnormality of an organ by using surgery is called an "invasive method"

* Invasive comes from the word "invasion", if we invade the body organ's to study it's abnormality etc by use of surgery — invasive method

Problem's associated with Invasive Method

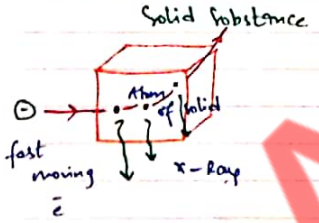
- Costly
- Time Consuming
- Painful
- Post operation infections

Non-Invasive Method

A process in which we study the structure, function or abnormality of an organ without using surgery or by using special techniques such as X-Rays, CT Scan, Ultrasound & pet scan is called "Non Invasive Method"

e.g.
- X-Ray
- CT Scan
- Ultrasound
- Pet Scan

X-Rays



* we have placed a solid substance made up of metal as they have high atomic No
* An electron moves towards the solid substance (fast moving) at it hit's the atom of solid substance & it decelerates and looses energy
* The lost energy appears as an electromagnetic radiation having short wavelength called "X-Ray"

* X-Ray's are produced as the λ / frequency of the electromagnetic radiation is unknown (other than ultraviolet, gamma or any other radiation)

* Similarly, the atoms also scatter as electron hits the atoms therefore the e^- hits neighbouring atoms and no of X-Ray's are produced but of different wavelengths as energy is lost by the e^- and speed of $\bar{e}$ also changes (decreases) (at different intervals)

X-Ray

When a fast moving electron moves towards a solid substance than it decelerates quickly and looses energy which appears in form of electromagnetic radiation having short wavelength and is called "X-Ray"

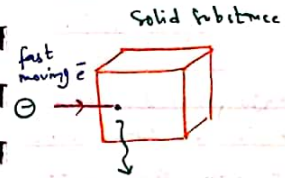
→ The λ of X-Ray's produced will vary but will be in the range of λ of X-Ray.

→ There are two type of X-Ray's
i- Continous X-Rays
ii- Characteristic X-Rays

* X-Ray's produced & has varying λ is called "Continous X-Ray" by deceleration of e^- while hitting a solid substance.

Cut off wavelength (λ_{min})

Minimum possible λ of X-Rays is called "cut off wavelength"



* we assume that $\bar{e}$ lost all it's energy after hitting the first atom of solid substance

Energy of X-Ray = Energy lost by electron

$$hf = W \quad \text{(work done by } \bar{e}\text{)}$$

$$f = \frac{c}{\lambda} \Rightarrow \frac{hc}{\lambda} = \Delta V \times q$$

$$\frac{hc}{\lambda} = \Delta V \times e$$

$$\boxed{\lambda_{min} = \frac{hc}{\Delta V \times e}}$$

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$\lambda = \frac{hc}{E}$ — energy

* as the energy is maximum (all energy lost ∴ max) so λ will be minimum

ΔV = Potential Difference of accelerating $\bar{e}$
e = charge on $\bar{e}$
c = speed of light
h = Plank's constant

→ Potential difference is required to accelerate the $\bar{e}$

* If we put all the values except ΔV then we will get the p.d at which $\bar{e}$ was accelerated to produce X-Ray

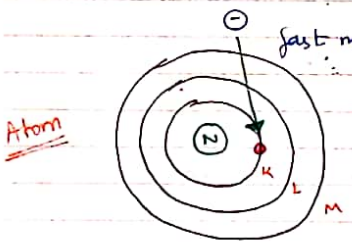
$$\Delta V = \frac{hc}{\lambda_{min} \times e}$$

* This is the method of producing X-Ray's (continous) by directing a fast moving $\bar{e}$ (produced by ΔV) towards a solid substance.

Continous X-Ray

X-Ray produced by the deceleration of $\bar{e}$ hitting the solid substance is called "Continous X-Ray"

↳ λ of X-Ray will always vary



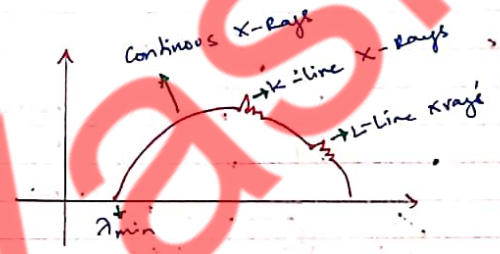
* $\bar{e}$ are placed in these subshell's of the atom of the solid substance
* There is a possibility that $\bar{e}$ can penetrate inside the subshell of the atom, knock's of an $\bar{e}$ in the atom thus a vacancy is produced to fill this vacancy, $\bar{e}$ from the higher subshell will jump to the lower subshell thus energy will be lost that will appear as electromagnetic radiation called "X-Ray"

↳ Characteristic X-Ray / line X-Ray

Characteristic X-Ray

X-Ray produced by the penetration of $\bar{e}$ inside the atom of solid substance is called "characteristic X-Ray" → λ of X-Ray also varies

* When we project a fast moving electron towards a solid substance, then both continous & characteristic X-Ray are produced.



Intensity of X-Ray

Number of X-Ray's produced per unit area per unit time is called "Intensity of X-Ray"

Penetrating Power

Ability of X-Ray to pass through a medium is called

* Depends on energy of X-Ray (Energy ∝ Penetration)

* With respect to penetrating power there are 2 type of X-Ray's;-
  i- Hard X-Rays
  ii- Soft X-Rays

Hard X-Rays

X-Rays of high penetrating power

Soft X-Rays

X-Rays of low penetrating power

* Soft X-Rays are dangerous as they can be absorbed by body which can cause mutations etc
  ↓
  cancer

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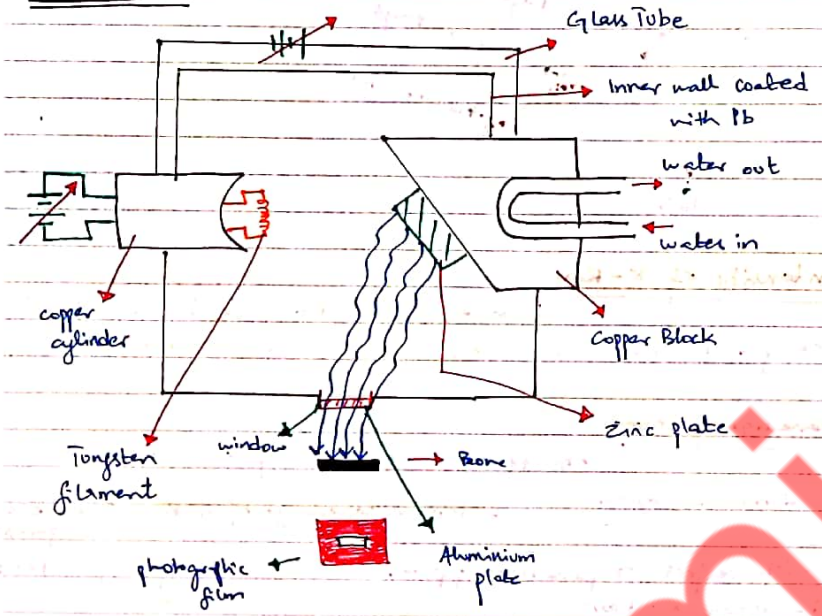
* X-Ray can pass through low density medium but can't pass through high density medium
  e.g
  bones will reflect the x-Ray's while organ won't

* When X-Rays hit the photographic film, it makes it dark

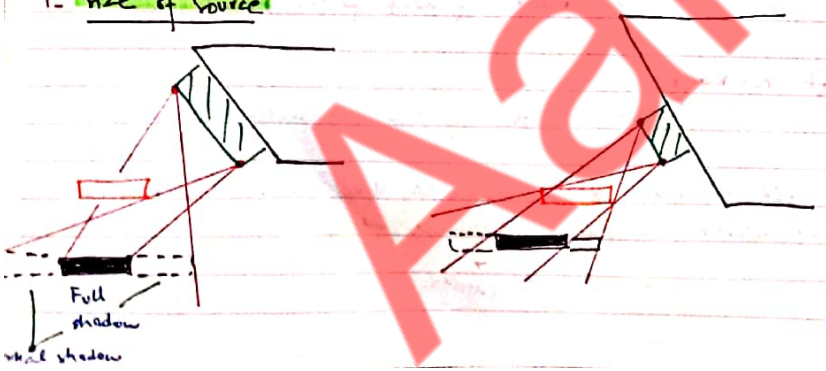
X-Ray Tube



Sharpness

(on 3rd page)

1. Size of Source



* As bone reflects the x-Ray, the region of bone will be bright as x-Ray can't pass through the bones, this will help us to identify the structure of bone etc
* We can produce x-Ray in a controlled fashion

X-Ray Tube

A device which produces X-Rays in a controlled way is called "X-Ray Tube"

→ Indirectly it means we can control the intensity & penetrating power of X-Ray

→ The inner wall of the glass tube is coated with Pb thus X-Rays can't pass through the Pb coating however only the window portion does not have Pb coating and has aluminium plate which blocks the soft X-Rays (as we need Hard X-Rays to avoid any medical emergency in the patient)

* Due to the battery the connected to the Cu cylinder, the cylinder will have -ve charge & Cu block will have +ve charge
* When we switch on this battery, heat is provided to the tungsten filament therefore e^- will be emitted by the filament which will be attracted towards the Cu block, due to this attractive force the e^- will accelerate towards the Zn plate
* When these e^- hit the Zn plate, X-Rays are produced which travel via the window towards the photographic film which becomes dark due to the X-Ray
* Bone is placed beneath the window and above the photographic film, X-Ray won't pass through the bone so the X-Rays will pass through the surrounding of bone and the shadow of bone will get dark (X-Ray hit's the film) so indirectly the shadow of bone is said to be X-Ray image while the bone region will be bright

* The battery at the left controls the intensity of the X-Ray while the battery at the top will control the penetrating power of X-Ray

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* x-Ray image's quality depends on sharpness of image and contrast of image

Sharpness

A property which gives the clear determination of the edges of the image is called "sharpness"

→ Sharpness depends on ;-
i. Size of source
ii. Size of window
iii. Scattering of x-Rays

→ Zn plate is the source of x-Ray
→ when x-Rays hit the bone, shadow of bone and partial shadow is also produced
→ Due to the partial shadow the edges are less sharp / less detectable
→ If we decrease the size of zn plate the partial shadow is reduced thus sharpness increases

Size of source	$\frac{1}{\propto}$	sharpness

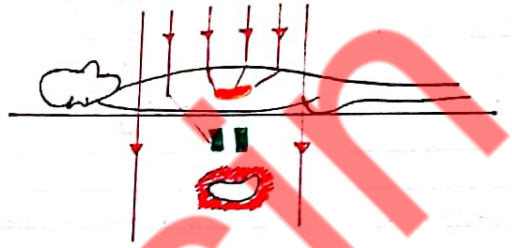
ii- Size of window

→ window is also indirectly a source of x-Ray thus it also is inversely related to the sharpness of x-Ray

Size of window	$\frac{1}{\propto}$	Sharpness

(on next page)

Scattering of x-Rays



* we are observing the stomach of the patient
* First we will give barium meal to the patient which contains $BaSO_4 + H_2O$
* x-Rays are passed through the body of patient $BaSO_4$ absorbs x-Rays thus the stomach region will stay bright on the photographic film as x-Ray don't pass through the stomach
* we want the surrounding region of stomach to hit the film for the x-Ray image to be produced
* Due to scattering of x-Ray, edges of image are not sharp, so we place a lead plate underneath the patient's stomach as it will absorb the scattered x-Rays

Contrast

Difference in the darkenings of the dark & bright areas of image is called "Contrast"

→ Contrast depends on ;-
i. Exposure Time
ii. Penetrating power
iii. Scattering of X-Rays

* The more the exposure time, the more x-Rays will pass and will make the dark area more dark along with a more brighter area

Exposure Time	$\propto$	Contrast

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* The more the penetrating power of x-ray, the more will be the contrast as more x-ray will pass and more darker area along with brighter area

$$\boxed{\text{Penetrating Power} \propto \text{Contrast}}$$

* The less the scattered x-rays the more will be the contrast

$$\boxed{\text{Scattered X-Ray} \quad \frac{1}{\alpha} \quad \text{Contrast}}$$

* I_0 is the intensity of x-ray before passing through medium and I is the intensity of x-ray after passing through the medium

μ = coefficant constant

Computed Tomography (CT scan).

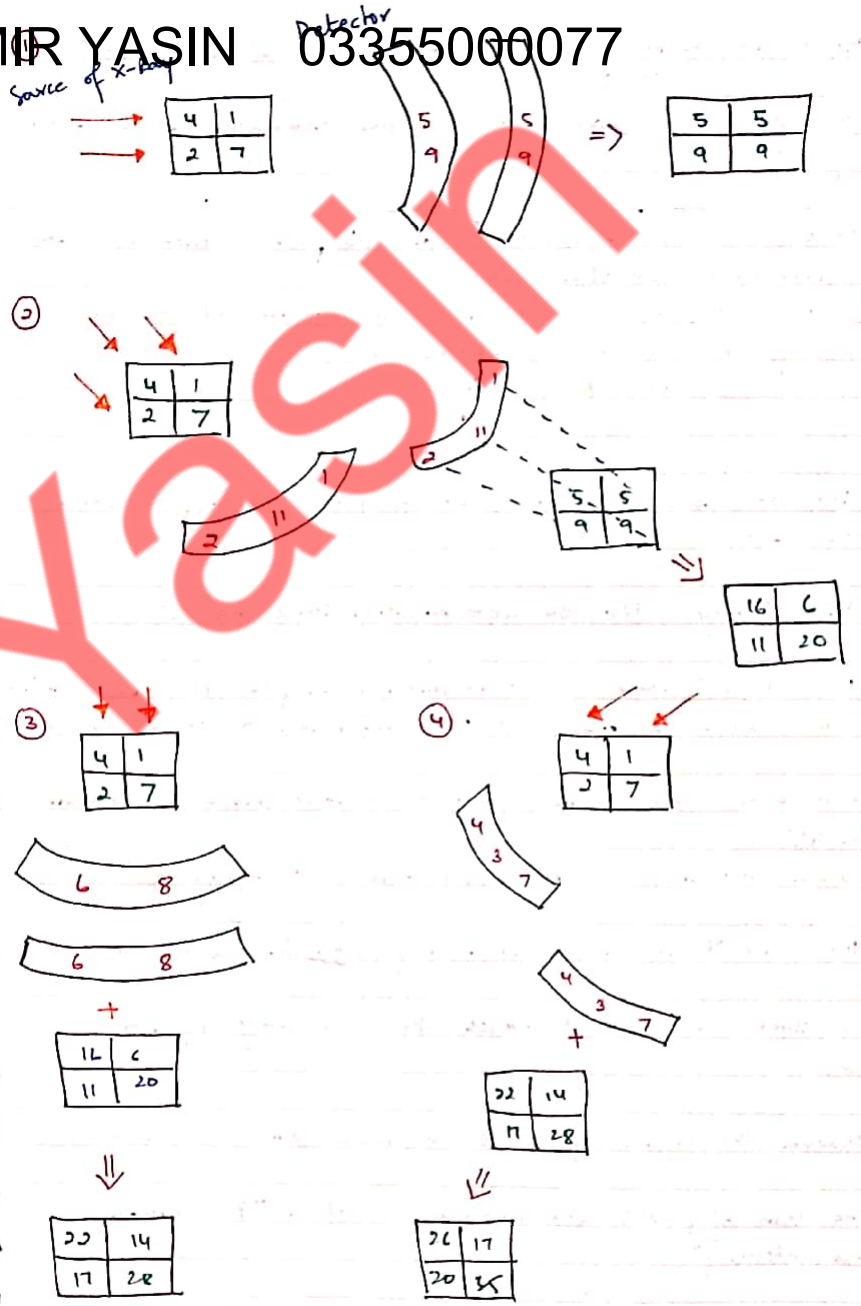
A process in which three dimensional image of an organ is formed using no of x-rays images and a computer programme

* X-Ray image is also called "Flat image" as it's 2-dimensional → Limitation of X-Ray image
* To study the structure of bone we would need a 3-D image
* we can use CT scan to make 3-D image of organ
* Take a no of x-ray images and combine them in a computer programme

Voxel / pixel

4	1
2	7

* The box is assumed to be selected section the organ to be studied
* we don't take CT scan of the complete organ rather we take 3-D images of sections of organ & combined them at the end



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* We divide the body into small pieces called "voxel"

* We assign each voxel with a certain intensity that is called "pixel".

* At one side of the selected sectional organ we place a source of x-ray and detector on the other side
* X-rays pass through the organ, then detector will get the image of the organ but intensities add up

eg. 4 + 1 = 5
2 + 7 = 9

* Detector passes the image to the computer and computer stores this image

* For 2nd image, turn the detector and source at 45°

* Now the 2nd image will also be stored in computer but now intensities will be added diagonally (as 2nd image added to the first image)

* After every image we will rotate the x-rays source & detector by 45°
* Again 3rd image will be added into 2nd image

* The same process is again repeated for the 4th image

* We assume that after 4 images the 3-D image of organ is made.

* However the original pixel is different from the 3-D image

* The sum of pixels set initially is called "Background Intensity"

OR

Sum of pixels produced on detector will be the background intensity

* No of images taken is called "Duplication value"

* Subtract the background intensity by the intensity of final image and then divide it by the duplication value
* In this way we will get the 3-D image with the same pixels as of initial

Voxel

The smallest part of the selected section of the organ is called "Voxel"

Background Intensity

26	17
20	35

14

12	3
6	21

Duplication Value

12	3
6	21

3

4	1
2	7

Pixel

Intensity of each voxel is called "Pixel"

Background Intensity

Sum of intensities on detector for each image is called "Background Intensity"

Duplication Value

No of images taken after first image is called "Duplication Value"

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Transducer

A device which converts one type of energy into another type of energy

Electric Transducer

A device that converts electrical energy in any other form of energy

Input Transducer

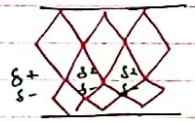
A device that converts any other form of energy into electrical energy is called "Input Transducer"

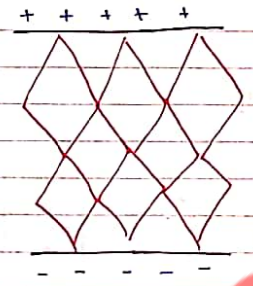
e.g) microphone

Output Transducer

A device which converts electrical energy in any other form of energy

e.g) speaker





* crystal is placed like this, and at it's normal state it won't be stretched or compressed. It has charges / e^- in the middle

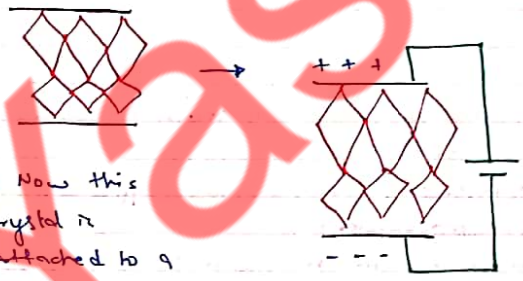
* Now we apply pressure on the crystal as it get's stretched that form as δ +ve & δ -ve charges at each end of crystal as electricity is produced, this process is called "Pressure electric effect"

* crystal is converting mechanical energy into electrical energy

↓

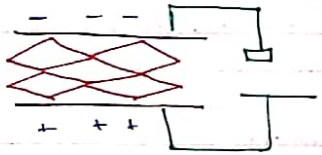
Act's as input transducer

* Now if we compress the crystal by applying pressure the -ve charge will be moved upwards & +ve to lower and again the charges are produced (electricity)



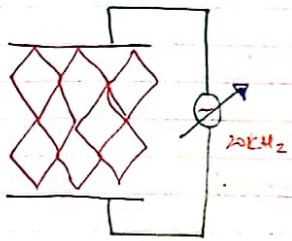
* Now this crystal is attached to a cell

=> crystal itself becomes stretched that upper end is connected to +ve terminal and lower end with -ve terminal



* It now becomes compressed as terminals are reversed

→ Now these crystal will convert electrical energy into mechanical energy thus acts as an output transducer



* Now the crystal is attached to A.C supply thus crystal will continuously stretch & compress

* If crystal placed in air, it will displace air particles to produce sound energy/wave

* If we keep 20kHz as frequency of A.C supply it will produce a sound wave of 20kHz called "ultrasound"

→ this is how ultrasound is produced

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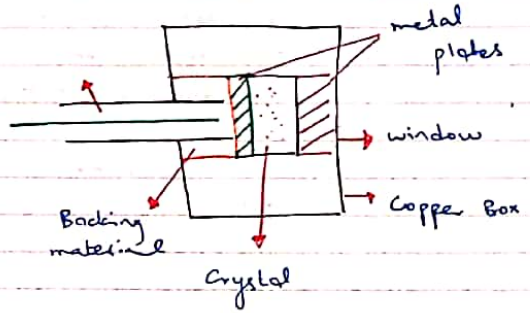
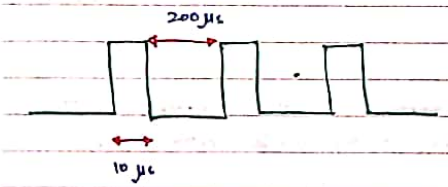
Piezo - Electric Transducer

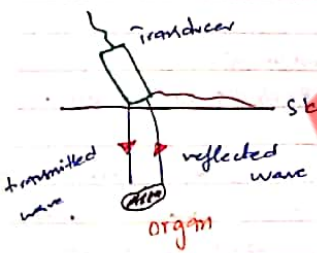


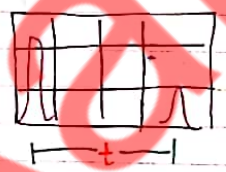
* Not only this arrangement produces ultrasound waves but also receive ultrasonic waves.
* we have placed transducer on skin (we first apply gel on skin and then place transducer)
* The transducer transmit's the ultrasonic waves and then receives the ultrasonic waves reflected by the organ
* Produces ultrasonic waves as pulses
* Acts as ~~receiver~~ transmitter for 10 μs and then for the next 200 μs acts as reciver.



* ultrasonic wave produced in form of short pulses

A - Scan





* We use ultrasonic waves for A & B-scan

A - Scan

A process in which depth of organ is found using ultrasonic waves is called "A - Scan"

* We have connected the transducer with a C.R.O, so when the ultrasonic wave is transmitted a pulse is formed on C.R.O and when ultrasonic wave reflects and transducer receives it another pulse is formed at the C.R.O

* We can find the time "T" b/w the 2 pulses, this time is the total time for the ultrasonic wave to be transmitted till organ and then back to the transducer

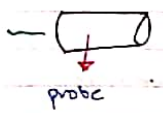
if we need time of wave to reach till organ only;-

$t = \frac{T}{2}$ $\quad S = vt$ $\quad$ v = speed of ultrasonic wave

S = Depth of organ ← S = distance covered by the wave to move from transducer to organ

B - Scan

A process in which a 2-Dimensional image of an organ is formed using an ultrasonic wave is called "B - Scan"



* image of unborn babies is → B-scan
* For B-scan we use probe
* Probe has no of crystals at single while transducer has only one crystal

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* Each crystal of probe will produce ultrasonic wave which will hit the tissue at different angles, the reflected waves will also be detected by probe
* A spot will be formed for every received wave flight

* The sequence of spots will gives us the 2-D image

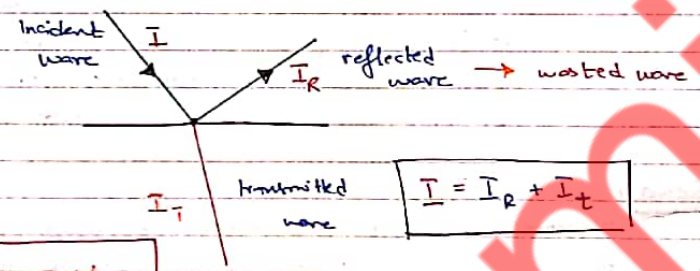
Specific Acoustic Impedance

Resistance offered by a medium to the flow of ultrasonic wave is called "Specific Acoustic Impedence"

$$Z = \rho c$$

ρ : density of medium
c : speed of ultrasound (in that medium)

* When ultrasonic wave moves from one medium to the other medium then the boundary of the medium reflects a certain amount of the ultrasonic wave



$$I = I_R + I_t$$

$$I_R = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2} I$$

* The more the difference of impedence of the mediums the greater will be the intensity of reflected wave
* For skin, air and fat are the 2 mediums
↳ The difference of impedence b/w air & fat is very much thus to avoid the reflection of wave we use gel having very similar impedence to skin → thus most of the wave passes through the skin

PET Scan (Positron Emission Tomography)

A pet scan creates 3-D images of the body. It does this by using radioactive tracers which are administered to a patient through intravenous injection. It is custom made for each patient

Tracers

→ The tracers are made up of carrier molecules that are tightly bound to a radioactive atom called an "isotope"

→ The carrier molecules can interact with or bind to specific proteins or organs in the body. The carrier molecule that will be used depends on what the detector is looking for

* Carrier molecules have different shape & structures depending on the organ to be scanned

→ The radioactive atom being unstable will emit radiation e.g in form of a B^+ — positron which will interact with e^-s of neighbouring tissues to form photons.

→ The isotope produces small particles called "positrons" which interact with the surrounding electrons, this interaction results in annihilation and two gamma rays photons are created which speed off in opposite directions

$$B^0_{+1} + B^0_{-1} \longrightarrow 2\gamma^0_0 + \text{Energy}$$

* photons travel in opposite directions to conserve momentum

→ The detectors in the PET scanner measures these photons & uses the information of thousands of these events every second and then processing the information through computing results in 3-D image.

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Annihilation of Mate-

$B^0_{-1} + B^0_{+1} \longrightarrow 2\gamma \rightarrow$ photons

$B^+ \xrightarrow{v} \quad \xleftarrow{v} B^-$

* B^+ & B^- travel with same speed but opposite direction

Initial $p = 0$
final $p = 0$

* To conserve momentum 2γ are produced

$$\boxed{K.E_{B^+} + K.E_{B^-} + E_{released} = 2E_\gamma}$$

* If v given then $K.E = \frac{1}{2}mv^2$ $\quad m = 9.11 \times 10^{-31}$ kg

* If speed not given then IGNORE the $K.E_{B^+} + K.E_{B^-}$

as it's assumed their v is very less than speed of light as $E_{released}$ is calculated by $e = mc^2$

if $v <<<<< c \longrightarrow$ K.E will be ignored

∴

Energy Released = $2E_\gamma$

$mc^2_{B^+} + mc^2_{B^-} = 2E_\gamma$ ∵ $\cancel{2} \times (9.11 \times 10^{-31})(3 \times 10^8)^2 = \cancel{2}E_\gamma$

$$\boxed{2mc^2 = 2E_\gamma}$$

$$\boxed{E_\gamma = 8.2 \times 10^{-14}\ J}$$

$E = hf \rightarrow$ To find out the frequency of photon

$f = 1.25 \times 10^{20}$ Hz (γ ray region)

* An important advantage of PET imaging is the fact that due to positron annihilation we expect to observe two photons at roughly the same time (in coincidence) in the detector ring.

↳ The annihilation event i.e the radioactive tracer will then be located somewhere on the line connecting the two photon detection points.

$$\boxed{d = c \times \Delta t}$$

$d \rightarrow$ location

* If position of tracer (annihilation) $\rightarrow \frac{d}{2}$ (Average)

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